

SIH26138: Executive Green Fleet Modernization & Decision Report

Maritime Decarbonization Intelligence | Generated: 2026-09-26 14:13:02 UTC

1. Executive Brief & Strategic Mandate

The proposed green fleet strategy transitions 1 vessels on the 3,500 nm corridor, achieving \$6.95M in annual operational savings (43.8%) and eliminating 4,822 metric tons of Well-to-Wake CO2e emissions (16.1%). With a dual-fuel retrofit capital allocation of \$0.0M across 0 assets, the plan delivers a projected 10-year ROI of 0.0% with a simple payback period of 0.0 years (ZERO_CAPEX). Crucially, schedule reliability is maintained at 100.0% through intelligent hydrodynamic eco-speed management (13.5 knots), ensuring full statutory compliance under both IMO MEPC.400(83) and EU FuelEU Maritime regulations.

2. Financial & Modernization Investment Economics

Metric	Value	Operational Rationale
Total Retrofit Capex	\$0.00M	Shipyard dual-fuel conversion allocation
Annual Net Benefit	\$6.95M	Net operational fuel and carbon tax savings
10-Year ROI	0.0%	Projected capital return over baseline
Payback Period	0.0 yrs	Status: ZERO_CAPEX
Annual Cost Savings	\$6.95M (43.84%)	Combined opex & regulatory penalty reduction
WTW CO2e Reduction	4,822 t (16.09%)	Lifecycle decarbonization impact

3. Operational Risk Mitigation Matrix

Operational Risk	Strategic Impact	Mitigation Strategy
Alternative Green Fuel Price Spikes	High opex inflation if green fuel market spread widens above \$140/t	Dual-fuel propulsion flexibility allows tactical switching back to MGO blended with HVO
Shipyard Retrofit Bottlenecks & Delays	Missed charter revenues and drydock overruns during engine overhaul	Staggered retrofit schedule to convert no more than 20% of fleet concurrently.
Regulatory Tightening (IMO MEPC.400(83) Phase 1 Acceleration)	Operational downgrade to D/E resulting in mandated speed reductions	Dynamic speed planning and wind-assist propulsion retrofit readiness.

4. Evidence vs. Assumption Classification

Item	Category	Regulatory / Physical Basis
IMO CII Reduction Targets (2026-2030)	STATUTORY	Adopted under IMO Resolution MEPC.400(83) (Z = 11.0% to 21.5% relative to 2019 baseline).
EU FuelEU Maritime Milestones (2025-2040)	STATUTORY	Legally enacted under Regulation (EU) 2023/1805 (2025: -2%, 2030: -6%, 2035: -14.5%, 2040: -31%).
Hydrodynamic Fuel Physics Formulation	MODELLED	Admiralty cubic power law verified against empirical sea trials and cross-source datasets.
Adverse Weather & Sea-State Degradation	ASSUMED	Modeled via Gaussian uncertainty distributions (sigma=0.08) calibrated to trade lane wave charts.